

1875

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BSSE23058.

Quiz-1

(CLO-1)-BT-C2

BSSE23 & BSCE22

Dr. Muhammad Asif

Human Computer Interaction

SE305T & MD445T (Spring-26)

Individual Assessment - Class

Information Technology University (ITU) - Lahore

27-Feb-2026

Total Marks: 20

Q1. List any four main elements of a computer system that affect interaction.

- Input devices
- Output devices
- Processing unit

(1.5)

Q2. Differentiate between input devices and output devices with one example of each.

Input devices are devices that are used to enter/input data. E.g.: Keyboard, mouse.

Output devices are devices that display the output after processing. E.g.: printer, monitor. ✓ (2)

Q3. Why is the QWERTY keyboard layout not considered optimal for typing?

QWERTY was used to avoid key jamming but it is not the best layout for typing as it is slow & inefficient. Dvorak is better. ✓

Dvorak

(2)

Q4. What is T9 predictive text entry? Write one advantage.

It is when an alphabet is typed and it suggests the next tent letter on its word. Fast data entry is the advantage. ✓ (2)

Q5. State two technical problems in handwriting recognition.

- Jagged & blurred out images. can. make handwriting detection harder. ✓

(2)

Q6. Differentiate between mechanical and optical mouse.

Mechanical mouse uses ball. Mechanical mouses are slow. & inefficient whereas optical mouses use light to ~~move~~ detect movement and is faster. (1.25)

Q7. Define resolution and colour depth in bitmap displays.

Resolution is the number of pixels.

Colour depth is the number of colors that are in the display. ✓

(2)

Q8. What is virtual memory? Why is it needed?

Virtual memory. is temporary memory like RAM. It is used to scale up memory when memory is needed. ✓

(2)

Q9. Differentiate between lossless and lossy compression with one example of each.

Lossless compression is when the size of a file is reduced without any data loss. eg ZIP files. Lossy compression is when data is cut down or deleted to reduce size. eg jpeg. ✓

(2)

Q10. List two limitations of networked computing in interactive systems.

Bottleneck problem. can be slow & inefficient as different networks. / devices are connected. processing power is slower & input-output feedback isn't fast due to batch processing. ✓

(2)

(9.5) + 0.5 = 10
Quiz-2
(CLO-1)

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Individual Assessment - Class

Information Technology University (ITU) -- Lahore

06-Mar-2026

Total Marks: 5 x 2 = 10

Q1. Differentiate between Direct Manipulation and Indirect Manipulation interaction styles.

- (2) Direct Manipulation is when user directly interacts with the screen, ~~either~~ by touching, like snapping a reel, clicking on something, computer responds immediately.
- Indirect Manipulation style is when user interacts with the computer through an intermediary, like example using Command lines, or keyboard shortcuts.

Q2. Briefly explain the role of rods and cones in the Human Visual System.

- (2) rods are used to see for night / dark vision, where colors don't matter but grayscale depth does. They are sensitive to light.
- cones are used for processing colors (RGB) in an image. it helps see the colors brightness, its depth. cones are for daylight vision.

Q3. What is Visual Modality in Human Computer Interaction? Explain how images or vision-based systems can be used as input to computers.

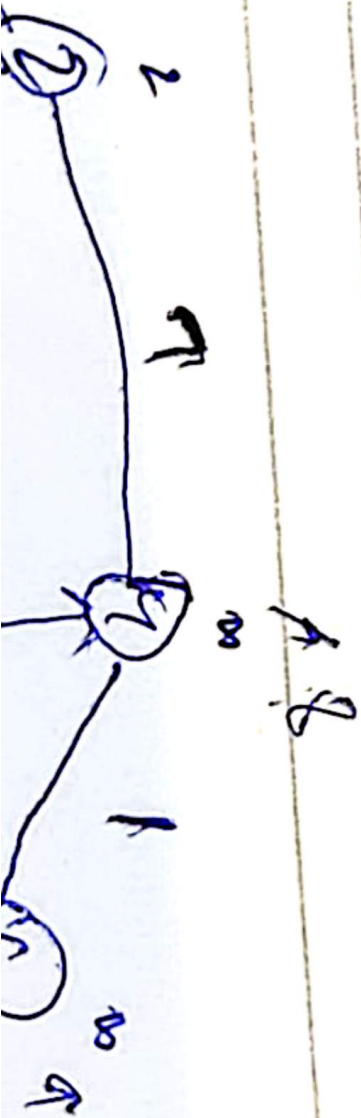
- (2) Visual Modality in HCI refers to the visual interaction of humans with computers. ~~this~~ examples are images, (digital pics) etc.
- Images can be used as input to computers by image acquisition, where ~~the~~ images are converted to digital images for processing.

Q4. Write any two applications of Image Processing in HCI systems and briefly explain them.

- (1.5 to 2) Image processing refers to cleaning the images like noise, ~~brity~~ brightness, ~~contrast~~ contrast etc.
- o OCR. - Optical Character Recognition - images are processed to extract machine readable text.
 - o Medical ~~image~~ screening - like ~~on~~ reading x-ray images.

Q5. State two type of image sources.

- (2) EM images like X-rays images.
Non EM images like ultrasound.



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Quiz-3

(CLO-2)

BSSE23 & BSCE22

Human Computer Interaction

SE305T & MD445T (Spring-26)

Individual Assessment - Class

Information Technology University (ITU) - Lahore

23-April-2026

Total Marks = $4 \times 2.5 = 10$

Q1. Design a simple conversation flow for a food ordering chatbot

User: Hi → Chatbot: Hi, would you like food?

A user inputs and asks something. The chatbot preprocesses the sentence (stopwords, punctuation, tokenization etc), Extracts features & sees the intent of the user and generates the best response according to the intent. Eg: "Hi, I would like to order food."

The chatbot cleans the text through NLP.

Q2. Differentiate between rule-based chatbot and AI-based chatbot in terms of design flexibility and user experience.

Rule based chatbot are bots that do not learn from the conversations. They are set up with pre defined structure and responses. Eg: FAQ bots.

Whereas, AI based chatbots learn from the data, they have a training & testing phase. They see the intent what the user is asking & generate the best response.

Q1. Text that is clean ["Hi", "Order", "Food"].

The chatbot sees the intent = Order-Food & generates the message.

"Hey, yes, tell me your order."
& it lists the menu.

Q3. Explain how tokenization and preprocessing affect chatbot performance.

Tokenization & preprocessing helps the chatbot's performance by cleaning the data and getting the intent more clearly. Tokenization extracts the text and ~~for~~ makes them into tokens to process them. Capitalization is removed so that 'Apple' & 'apple' are treated the same way. Removing stop words helps get the main message or intent of the user more clearly. But it can affect the performance sometimes too. Like removing stop words could potentially like I ~~don't~~ like you. Sometimes 'don't' is removed and the intent gets ruined.

Q4. Explain the role of training data in designing an NLP-based chatbot.

Training data's role in designing the NLP helps in getting ^{me} better ~~be~~ out of chatbot so that it doesn't fail in production. Training the chatbot with datasets improves its accuracy to handle different cases.

2.5